Session 1: AI for Coding — Tools, Concepts, and Setup

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The George Washington University
May 4, 2026

What We Will Cover (Sessions 1 & 2)

- Part 1: Tools, concepts, and setup
  - Overview of AI coding tools and how to choose among them
  - How to use AI Agents: Terminal, Desktop App, VS Code
  - Conceptual overview of context engineering with Claude
  - VS Code and Claude Code integration
  - Git and Gihub: why they matter for research and Claude

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  - Conceptual overview of context engineering with Claude
  - VS Code and Claude Code integration
  - Git and Gihub: why they matter for research and Claude
- Part 2: Practical workflows
  - Writing, debugging, and automating code with AI
  - “Vibe coding” from plain English descriptions
  - Project organization and replication packages
  - Demo: data cleaning script from scratch (python and stata)

Coding Perhaps Highest-Value Use of AI

- AI most reliable when output is verifiable: code runs or it does not
- Economists spend enormous time on repetitive coding tasks
  - AI can handle most of them
- Massive productivity gains across skill levels:
  - Beginners: “vibe code” to write code in an unfamiliar language (no syntax!)
  - Advanced users: automate drudge work, prototype faster, reduce bugs

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$\Rightarrow$ First two sessions will cover how to use AI to complete coding tasks

Overview of AI Coding Tools

The AI Coding Landscape: Many Options

- General-purpose AI assistants (good at coding, among many other things):
  - **Claude** (Anthropic) — our focus today
  - **ChatGPT / GPT-4o** (OpenAI)
  - **Gemini** (Google)
- Agentic coding tools (read files, write and run code autonomously):
  - **Claude Code** (Anthropic) — our focus today
  - **Codex** (OpenAI) — closest competitor to Claude Code
  - **GitHub Copilot** (Microsoft) — inline suggestions in editor
  - **Cursor / Windsurf** — AI-first code editors
- How to choose: task type, privacy needs, and workflow integration

Chatbot vs. Agentic Tool

- **Chatbot** (Claude.ai, ChatGPT): you ask → it answers → *you* act
- **Agentic tool** (Claude Code): you give a goal → it reads files, writes code, runs it, fixes errors, iterates → reports back
- For economists: instead of "here is some code," it runs the code and tells you what it found

Which One Should You Use? Claude Code vs. Codex

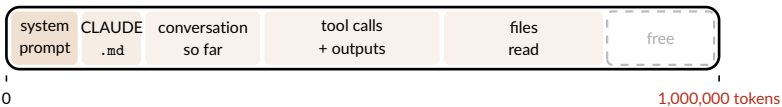
- Two serious options for agentic coding today: **Claude Code** and **Codex**
- Plans mirror each other at each price point:
  - $20/mo: Claude Pro vs. ChatGPT Plus
  - $100/mo: Claude Max (5x) vs. ChatGPT Pro (5x)
  - $200/mo: Claude Max (20x) vs. ChatGPT Pro (20x)
- Both use a 5-hour rolling window + weekly cap; limits reset automatically
- At the same price, Codex currently gives more usage per dollar

Which One Should You Use? Claude Code vs. Codex

- Two serious options for agentic coding today: **Claude Code** and **Codex**
- **Codex** has some advantages:
  - More usage per subscription dollar — important for heavy users
  - Sometimes stronger on pure coding benchmarks
- **Claude Code** has some advantages:
  - Often better at tasks beyond code: understanding context, writing, reasoning
- **Bottom line:** this comparison is fast-moving and will keep changing
  - I am using Claude Code and find it excellent for my workflow
  - I would switch only if Codex proved clearly and consistently superior
  - My advice: pick one, learn it well, reassess in 6 months

What You Pay For: Tokens and Context

- **Token**: the unit AI reads and writes — roughly $\frac{3}{4}$ of a word
  - **You pay for**: tokens *in* (prompt + files) and tokens *out* (reply) — input is cheaper
- **Context window**: everything the model holds at once — fills as session grows



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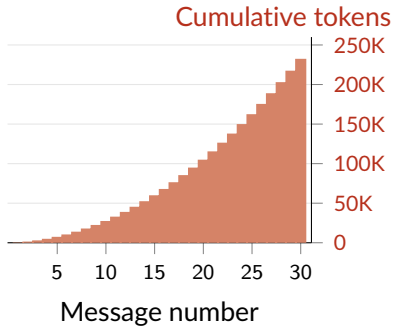
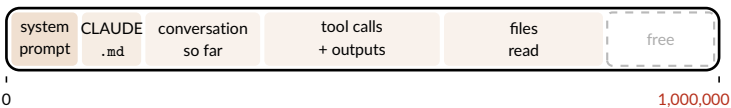




Getting the Most Out of Your Subscription

- Subscriptions have rolling limits — fewer tokens per task means more use
- How to use fewer tokens:
  - Start a new session for each new task: context accumulates and raises costs
  - Store standing instructions once so you never repeat yourself
  - Be specific: vague prompts waste tokens and invite unfocused replies
  - Ask multiple things at once so less back and forth
  - Compact your chats/context window
- Silly trick: start Claude when wake up so 5-hour window ends earlier in day

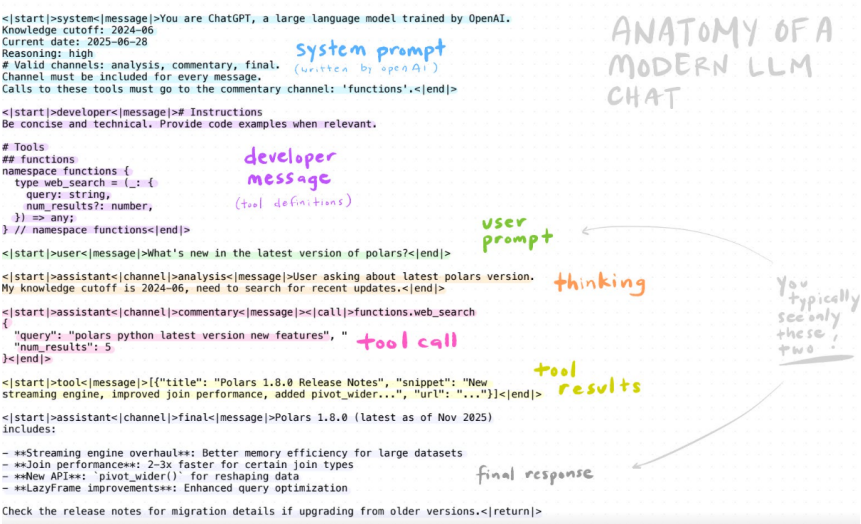
Contex Engineering with Claude Overview

Contex Engineering

- Context Window
  - Each prompt to Claude Code bundles the entire history of your session
    - your prompts, Claude's responses, files it read, code outputs, system promtsz
  - These are all appended until the context window hits it's fixed size limit
    - For Claude Code this is around 200k tokens
  - Claude then "**auto-compacts**" writes it's own summary and then restarts fresh
    - less preferable to frequent intentional compaction
- Contex Engineering
  - You can tell claude when to compact using `/compact your instructions`
  - Develop habits that the context window focussed on relevant information
    - Claude.md vs skills —and— 'ask before edits' vs 'plan' Modes

Read more at Paul GP's Guide

Anatomy of a context window (example from Paul GP's Guide)



<|start|>system<|message|>You are ChatGPT, a large language model trained by OpenAI.
Knowledge cutoff: 2024-06
Current date: 2025-06-28
Reasoning: high
Valid channels: analysis, commentary, final.
Channel must be included for every message.
Calls to these tools must go to the commentary channel: 'functions'.<|end|>

<|start|>developer<|message|># Instructions
Be concise and technical. Provide code examples when relevant.

Tools
functions
namespace functions {
  type web_search = (_: {
    query: string,
    num_results?: number,
  }) => any;
} // namespace functions<|end|>

<|start|>user<|message|>What's new in the latest version of polars?<|end|>

<|start|>assistant<|channel|>analysis<|message|>User asking about latest polars version.
My knowledge cutoff is 2024-06, need to search for recent updates.<|end|>

<|start|>assistant<|channel|>commentary<|message|><|call|>functions.web_search
{
  "query": "polars python latest version new features", "
  "num_results": 5
}<|end|>

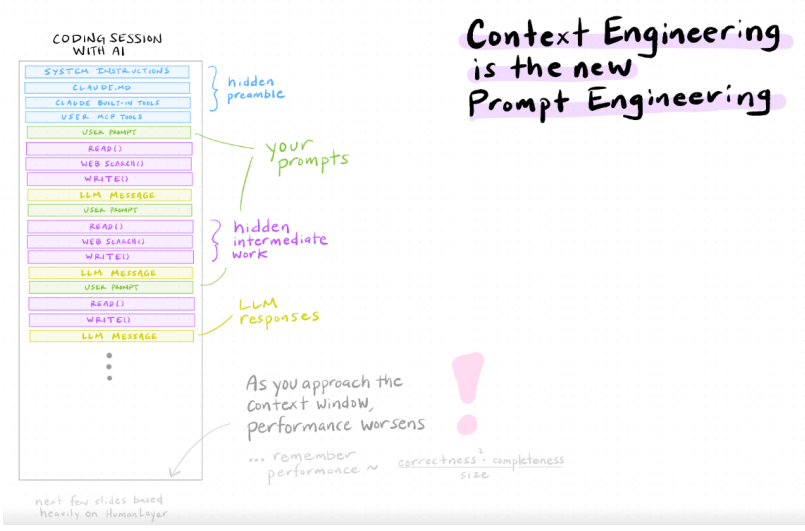
<|start|>tool<|message|>[{"title": "Polars 1.8.0 Release Notes", "snippet": "New
streaming engine, improved join performance, added pivot_wider...", "url": "..."}]<|end|>

<|start|>assistant<|channel|>final<|message|>Polars 1.8.0 (latest as of Nov 2025)
includes:

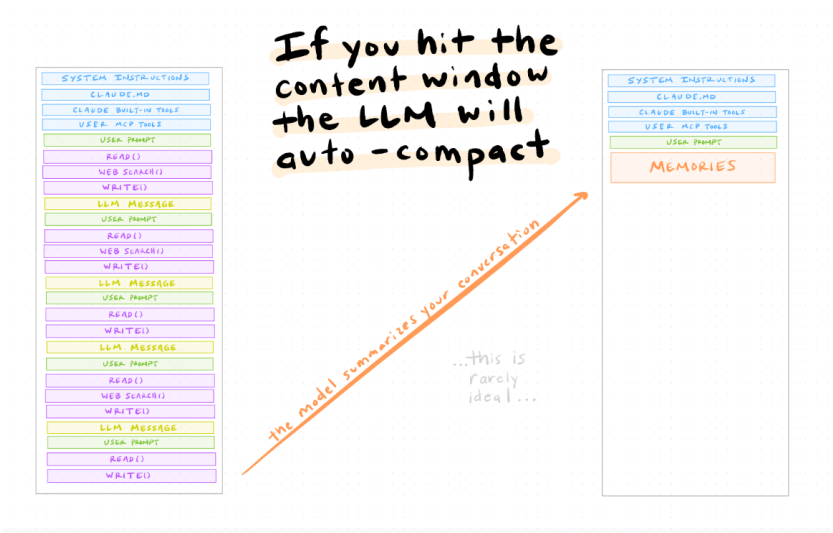
- **Streaming engine overhaul**: Better memory efficiency for large datasets
- **Join performance**: 2–3x faster for certain join types
- **New API**: `pivot_wider()` for reshaping data
- **LazyFrame improvements**: Enhanced query optimization

Check the release notes for migration details if upgrading from older versions.<|return|>

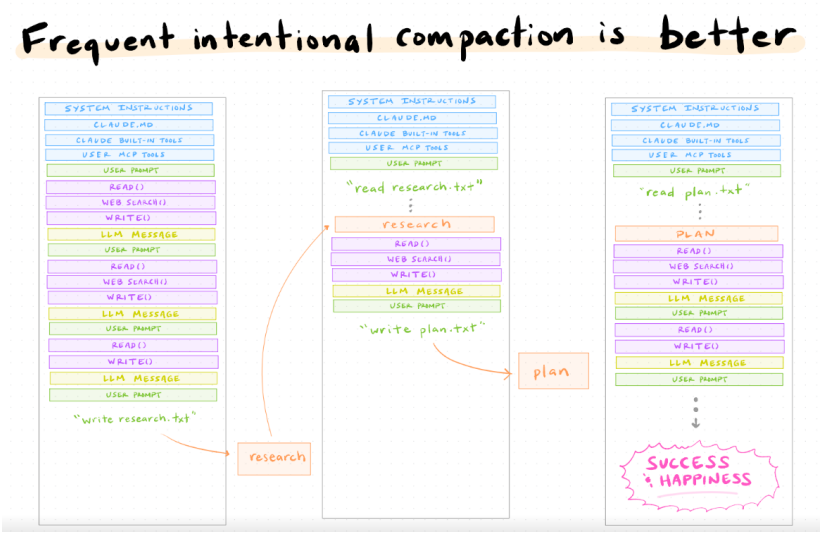
Context engineering (example from Paul GP's Guide)



Context auto-compact (example from Paul GP's Guide)



Context intentional compacting (example from Paul GP's Guide)



CLAUDE.md and Skills

- **CLAUDE.md** (memory) — Claude reads at the start of every session
  - Store context that is always true so you don't repeat yourself in every prompt
  - `.claude/CLAUDE.md` is global, `your-project/CLAUDE.md` is *project* specific
  - Ask Claude to update after substantial changes to project
- **Skills** — Claude reads every time it is called
  - Store context for a task that you repeat often, but not always relevant
  - `.claude/skills/your-skill/SKILL.md` is called with `/your-skill`
  - Write and update your own skills or download from others

Read more at Claes Backman's Guide (full descriptions, examples of CLAUDE.md and skills, download pre-made skills)

`your-project/CLAUDE.md` (example from Claes Backman's Guide)

```
# Project: [Paper title]

## Data
- Unit of analysis: firm-year panel, 2010–2022
- Raw data lives in data/raw/ — never edit these files
- Main dataset after cleaning: data/clean/panel.dta (N ≈ 47,000)

## Code conventions
- R scripts are numbered (01_clean.R, 02_analysis.R, ...)
- All regressions cluster SEs at the firm level
- Use fixest for all panel regressions

## LaTeX
- Main file: paper/main.tex
- Tables generated by 03_tables.R go to paper/tables/
- Use \widehat{} not \hat{} for estimators
```

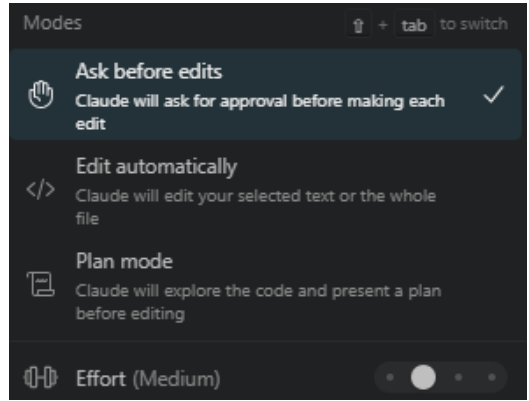
`.claude/skills/robustness/SKILL.md` (example from Claes Backman's Guide)

```
---
name: robustness
description: Run standard robustness checks on the main regression.
---

For the main specification in this project:
1. Re-run with alternative clustering (industry-year instead of firm)
2. Re-run dropping the top and bottom 1% of the outcome variable
3. Re-run on the pre-2020 subsample only
4. Produce a summary table comparing coefficients across all specs
```

Claude Modes

- **Ask before edits**
  - Do steps one at a time
  - Asks your permission before running each and every command
- **Edit automatically**
  - Same as above but runs commands without asking permission (risky)
- **Plan**
  - Claude writes plan.md with many steps
  - You review and execute entire plan



Other tips for managing the context window

- **Use @-mentions to be selective**
  - Rather than Claude reading everything, point to a specific file
  - e.g. `update all the pathnames in @Lecture_2.tex`
- **Check usage with /context**
  - Type /context in the panel to see a breakdown of how your context window is being used and how much space remains.

Claude Code / Codex Interfaces

Three Ways to Use an Agentic Coding Tool

- **Terminal** (command line):
  - Most powerful; works directly in your project directory
  - Best for: running scripts, data analysis, replication tasks
- **Desktop App**:
  - Friendlier interface; easier to get started
  - Projects feature: persistent context across sessions
  - Best for: drafting, quick questions, early exploration
- **VS Code integration**:
  - Agent lives inside your editor — see code and AI side by side
  - Best for: sustained development work, Git integration, Jupyter/Python

Terminal vs. VS Code: Quick Comparison

Terminal (CLI)

- Lightweight; fast startup
- Works anywhere, incl. remote servers
- Keyboard-driven; fast for quick tasks
- No inline diffs; steeper setup

VS Code Extension

- Inline diffs; accept changes block-by-block
- Code & AI panel side by side
- Best for large codebases, sustained work
- Heavier on CPU and memory

Live Demo: Terminal

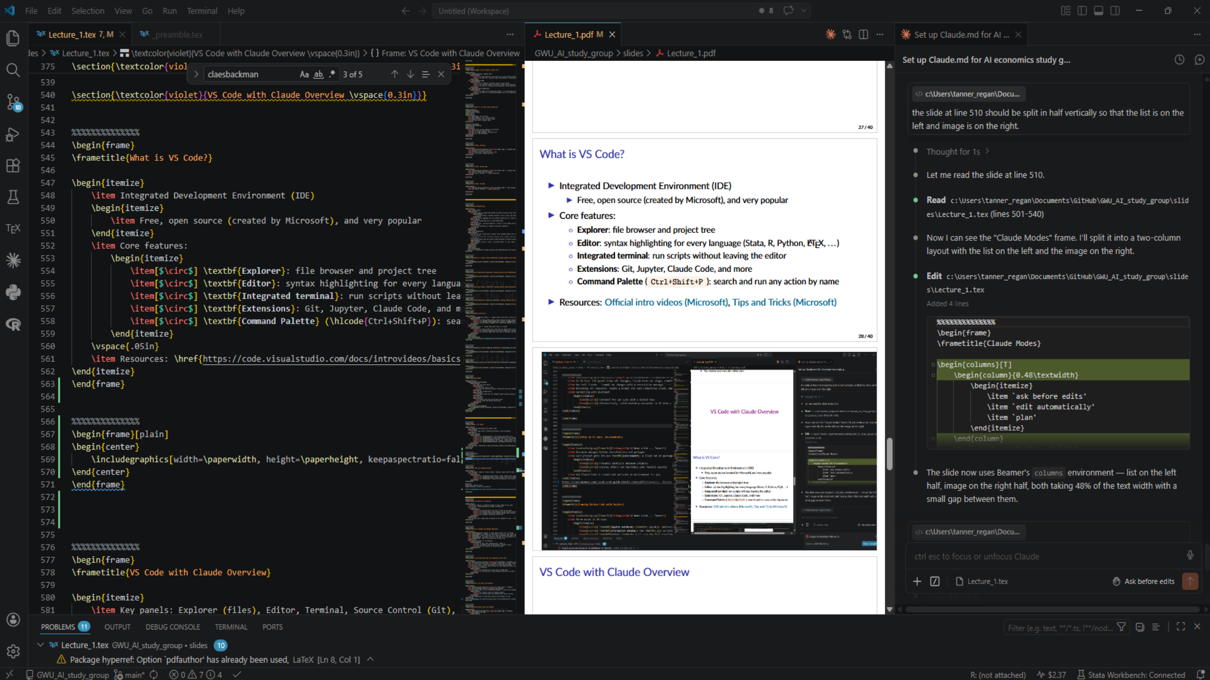
Live Demo: Desktop App

Live Demo: VS Code

VS Code with Claude Overview

What is VS Code?

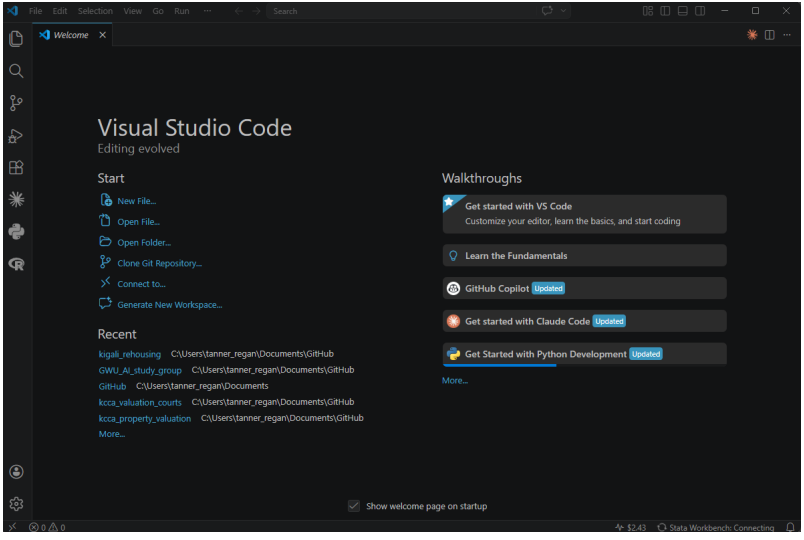
- Integrated Development Environment (IDE)
  - Free, open source (created by Microsoft), and very popular
- Core features:
  - **Explorer**: file browser and project tree
  - **Editor**: syntax highlighting for every language (Stata, R, Python, LaTeX, …)
  - **Integrated terminal**: run scripts without leaving the editor
  - **Extensions**: Git, Claude, Stata, latex, python, etc. Starter Pack for Economists.
  - **Command Palette** (`Ctrl+Shift+P`): search and run any action by name
- Claes Backman's Guide Claude Code in VS Code for Academic Economists
  - Other resources: Official intro videos (Microsoft), Tips and Tricks (Microsoft)


File Edit Selection View Go Run Terminal Help
Untitled (Workspace)
Lecture_1.tex 7, M
_preamble.tex
Lecture_1.pdf M
Set up Claude.md for AI ...
\section{\textcolor{violet}{VS Code with Claude Overview \vspace{0.3in}}}
%%%%%%%%%%%%%%
\begin{frame}
\frametitle{What is VS Code?}
\begin{itemize}
\item Integrated Development Environment (IDE)
\begin{itemize}
\item Free, open source (created by Microsoft), and very popular
\end{itemize}
\item Core features:
\begin{itemize}
\item[$\circ$] \textbf{Explorer}: file browser and project tree
\item[$\circ$] \textbf{Editor}: syntax highlighting for every langua
\item[$\circ$] \textbf{Integrated terminal}: run scripts without lea
\item[$\circ$] \textbf{Extensions}: Git, Jupyter, Claude Code, and m
\item[$\circ$] \textbf{Command Palette} (\hlcode{Ctrl+Shift+P}): sea
\end{itemize}
\vspace{.05in}
\item Resources: \href{https://code.visualstudio.com/docs/introvideos/basics
\end{itemize}
\end{frame}
%%%%%%%%%%%%%%
\begin{frame}[plain]
\begin{center}
\includegraphics[width=\paperwidth, height=\paperheight, keepaspectratio=fal
\end{center}
\end{frame}
%%%%%%%%%%%%%%
\begin{frame}
\frametitle{VS Code with Claude Overview}
\begin{itemize}
\item Key panels: Explorer (files), Editor, Terminal, Source Control (Git),
claesbackman
3 of 5
GWU_AI_study_group > slides > Lecture_1.pdf
27 / 40
What is VS Code?
Integrated Development Environment (IDE)
Free, open source (created by Microsoft), and very popular
Core features:
Explorer: file browser and project tree
Editor: syntax highlighting for every language (Stata, R, Python, LaTeX, ...)
Integrated terminal: run scripts without leaving the editor
Extensions: Git, Jupyter, Claude Code, and more
Command Palette (Ctrl+Shift+P): search and run any action by name
Resources: Official intro videos (Microsoft), Tips and Tricks (Microsoft)
28 / 40
VS Code with Claude Overview
Set up Claude.md for AI economics study g...
c:\Users\tanner_regan\Docu...
the slide at line 510 should be split in half vertically so that the list is on the left and image is on the right.
Thought for 1s
Let me read the slide at line 510.
Read c:\Users\tanner_regan\Documents\GitHub\GWU_AI_study_group\slides\Lecture_1.tex (lines 501-540)
Now I can see the "Claude Modes" frame. I'll split it into a two-column layout with the list on the left and the image on the right.
Edit c:\Users\tanner_regan\Documents\GitHub\GWU_AI_study_group\slides\Lecture_1.tex
Added 4 lines
%%%%%%%%%%%%%%
\begin{frame}
\frametitle{Claude Modes}
\begin{columns}[T]
\begin{column}{0.48\textwidth}
\begin{itemize}
\item `ask before edits'
\item `edit automatically'
\item `plan'
\end{itemize}
\end{column}
The slide now uses Beamer's columns environment — list on the left half, image on the right half, both taking 48% of the text width with a small gap between them.
c:\Users\tanner_regan\Docu...
ctrl esc to focus or unfocus Claude
Lecture_1.tex
Ask before edits
PROBLEMS 11 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Lecture_1.tex GWU_AI_study_group • slides 10
Package hyperref: Option `pdfauthor' has already been used, LaTeX [Ln 8, Col 1]
GWU_AI_study_group main*
R: (not attached) $2.37 Stata Workbench: Connected

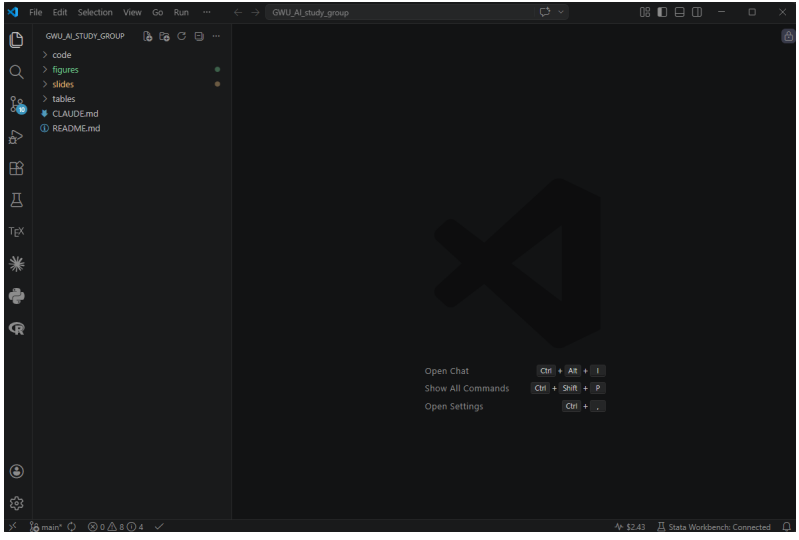
Claude Code **extension** in VS Code

- Give Claude access to your whole project directory to maximize its usefulness
  - Open your full project folder in VS Code (File → Open Folder)
  - This can be any folder on your computer — plain folder, a git repository, a Dropbox folder, etc.

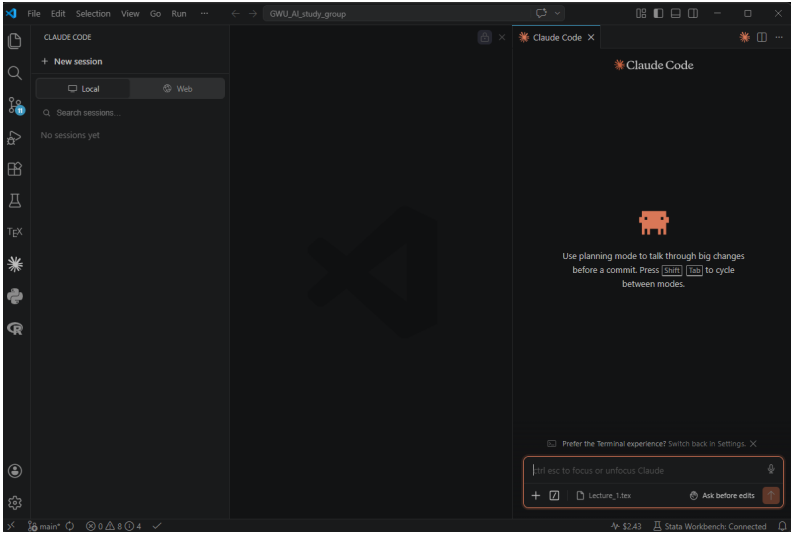
New VS Code Window — open new folder



Project folder is open — now open a new Claude Code session

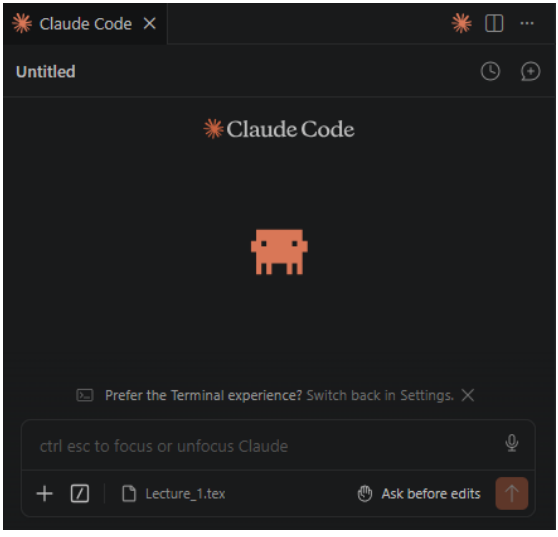


New Claude Code session is open



Claude Code **extension** in VS Code

- Each new conversation in the Claude panel is a fresh context
  - Click the + button at the top of the Claude panel to start one,
  - or type `/clear` to reset
- Go over a fresh window $\longrightarrow$
  1. Session Title
  2. Session History
  3. Mode menu
  4. Current File Selection
  5. Attachments
  6. Command Menu



Files and Data that Claude is good at reading

- Claude Code works best with plain text files
  - e.g. `.txt`, `.do`, `.py`, `.tex`, `.csv`, `.log`, `.md`, etc.
- Tips for optimial performance with other file types
  - `.docx`: convert to markdown with `pandoc file.docx -o file.md`
  - `.xlsx`, `.dta`, `.rds`, `.sav`: convert to CSV first with a short script
  - `.pdf`:
    - Use the `.tex` file whenever possible (especially if equations and tables)
    - Long PDFs (50+ pages): convert to markdown first
    - Scanned PDFs: run OCR first, then convert to markdown
- Create a skill that loads file types, e.g. `/pdf-to-markdown`

Read more at Claes Backman's Guide (details on handling different file types)

Git: Version Control for Coding Projects

Why Git Matters for Research

- Version control = a complete history of every change to your code
  - No more: `analysis_v3_FINAL_revised2.do`
- Enables:
  - Backup: retrieve any prior version of your code
  - Collaborate: coauthors and RAs edit without overwriting each other
  - Track: what changed and who made the changes at every stage
- Can help improve project replicability both internally and externally
  - an automatic audit trail
  - sharing and iterating on projects publicly

Read more at Luís Fonseca's Version Control with git (for economists)

Git and Github Concepts

- Key concepts to remember
  - **Repository (repo)**: a folder whose history Git is tracking
  - **Commit**: a snapshot of the code at a point in time, with a descriptive message
  - **Sync**: Pushes and pulls at the same time, so both repos are up to date
- Slightly more advanced concepts (learn these when you need them)
  - **Push**: sends changes up from your local repo to online/remote repo
  - **Pull**: brings changes down to your local repo from online/remote repo
  - **Branch**: a parallel version of the code (useful for exploration and RA work)
  - **Merge**: fold a branch back into the main code
  - **Pull request**: propose to merge one branch to another (e.g. main)
  - **Clone**: copy an online repo to local computer for initial setup
  - **Fork**: copy an online repo to your own GitHub account

Git vs. GitHub: Two Separate Things

git — the software that runs locally

- All history lives in the `.git` folder
- Records *commits*, *branches*, *merges*
- Compares any two versions line-by-line; flags conflicts

GitHub — online hosting platform

- Stores a remote copy of your repo
- Alternatives: GitLab, Bitbucket
- Remote endpoint for *Push*, *Pull*, *Clone*; introduces *Fork*

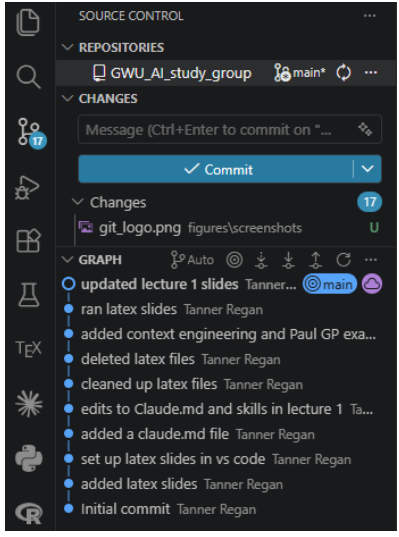
Typical Git workflows for an economist

- Typical workflow for an economist (alone or with trusted RA)
  1. Pull/Sync code at start of work session (get latest version of online repo)
  2. Work on code; when something works, *commit* it with a descriptive message
  3. Push/Sync at end of work session (updates the online repo)

- More elaborate, with explicit RA review
  1. RA syncs and then creates a new branch
  2. RA works on branch, makes commits and makes a pull request
  3. You review the pull request (compare the branch to main)
     - recall git highlights differences line by line and flags conflicts
  4. You merge back into *remote* main branch, or ask RA for fixes
  5. Pull/Sync to update your local repo

Git source control panel in VS Code

- Source Control panel
  - Click the github icon to open
- Go over an example $\longrightarrow$
  1. Icon with number of changes
  2. Message bar for commit
  3. Pull/Sync button
  4. commit button (also push)
  5. List of changes
  6. Graph of commits and branches



Setting up a github repository

- Go to github.com
  - Create an account, add new repository, add collaborators
  - Create a readme file that describes the project
- Open VS Code git extension (and sign into your GitHub account)
  - clone the repo to local computer: `Ctrl + Shift + P` (Windows/Linux) or `Cmd + Shift + P` (Mac) to open the Command Palette. Type `Git: Clone` and press Enter. Choose from the list of your repos.
- I have created a GitHub repo for our workshop GWU_AI_study_group
  - It is setup as Public, so you can already look at it and even Fork it
  - If you make a GitHub account and send me your username, I will add you as a collaborator so you can push changes to the repo as well

What Goes in a Repository?

- **Do NOT** store any datasets (source or generated)
  - instead store in Box (GW-endorsed secure storage)
  - or locally, Dropbox, Google Drive, etc.
    alternatively, write a `.gitignore` file that tells git to ignore specific subfolders (e.g. `/data`)
- **Do** store all other project materials
  - code (Stata, Python, LaTeX)
  - output: tables, figures
  - CLAUDE.md, notes, slides

```
GWU_AI_study_group/
├── slides/
│   ├── _preamble.tex
│   ├── Lecture_1.tex
│   └── Lecture_2.tex
├── code/
├── figures/
├── notes/
├── tables/
├── CLAUDE.md
└── README.md
```

Git and Claude Code Integration

- Claude Code is get-aware
  - When the open project is a git repo, Claude can see the entire commit history

- Claude can facilitate git operations
  - Write commit messages `commit my changes with a descriptive message`
  - Explain what changed `what changed between commit A and B?`
  - Resolve a merge conflict `fix RA code to merge with the main branch`

Read more at Claes Backman's Guide

Setting Up Your Environment

Core tools needed to set yourself up

- Software to install (first two are substitutable)
  1. **VS Code** — free to download here
     - + Claude Code extension
  2. **Claude Code** — free to download here
  3. **Git** — free to download here
- Subscriptions and accounts to create
  1. **Claude Code** — subscribe for a fee (here)
  2. **GitHub** — free account (here), and send Tanner your username!
- Guides
  - A step-by-step setup guide for Mac (here) and Windows (here)
  - However, Claude can walk you through any setup as long as you:
    - install VS Code + Claude Code estension
    - or install Claude Code desktop app

Other set up if you want to follow along next class

- Software to install (first two are substitutable)
  1. **Miniconda** — free to download here, python will come included
  2. **MikTex** — free to download here, I also had to install Strawberry Perl
  3. **Stata** — our old friend

- VS Code Extensions
  - Claude Code, LaTex Workshop, Stata Workbench, Python, Jupyter
  - See also Claes Backman's Starter Pack for Economists

Session 1 — Key Takeaways

1. AI has perhaps most value added in coding — output is verifiable
2. **Agentic** $\neq$ **chatbot**: reads files, writes and runs code, fixes errors, reports back
3. Two serious options today: **Claude Code** and **Codex** — pick one
4. **Manage your context window**: tokens accumulate fast — start fresh sessions, compact often, use CLAUDE.md so you never repeat yourself
5. **Git + GitHub** are your safety net — commit before big changes, push often
6. Three interfaces (Terminal, Desktop, VS Code) — VS Code integrates everything

Questions?

- Questions on tools, concepts, or setup?
- Thoughts on how this fits your current workflow?